

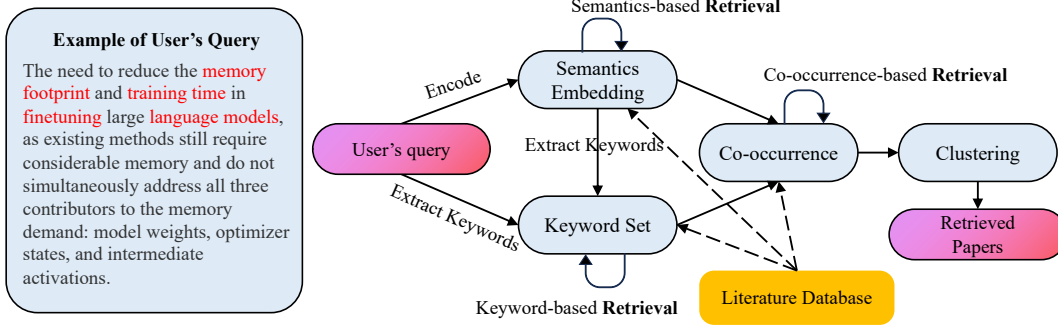


Figure 2. The pipeline of SKC-based literature retrieval and literature clustering. Red words in the user’s query are entity examples.

we record all co-occurrence relationships of papers and keywords. As illustrated in Figure 1, if a keyword T_{e1} is mentioned in paper p_1 , an edge is created between the corresponding paper and keyword nodes. This structure facilitates efficient graph-based traversal and retrieval.

3.2. Multi-granularity Literature Retrieval

Based on our constructed database, we present a multi-granularity literature retrieval framework that integrates three dimensions: keywords, semantics, and citation relationships. Notably, SciPIP does not simply aggregate the retrieval results from keywords, semantics, and citations. Instead, it introduces SKC-based (meaning semantics, keyword, and co-occurrence) retrieval approach that allows the three dimensions to leverage information from one another, thereby enhancing the overall retrieval performance, as illustrated in Figure 2. Furthermore, to mitigate redundancy in the retrieval results, we apply a clustering method to group papers with similar ideas, ensuring that the final retrieval results are free from highly similar references.

Semantics-based Retrieval. As shown in Figure 2, during the retrieval process, the user’s query $T_b^{(u)}$ generally includes the background of a research field. We first encode the query as $e_b^{(u)}$ using the same embedding model as constructing the database and then compare it with the background embeddings of the papers stored in our database. This approach enables the retrieval of literature that addresses similar challenges, providing more targeted and relevant results. Formally, let this subset of retrieved papers be denoted as $\mathbb{N}_1$,

$$\mathbb{N}_1 = \{p | e_b^{(p)} \in \text{TopK}(\text{cosine}(e_b^{(u)}, e_b^{(i)})) \text{ for } i \in \mathbb{D}\}, \quad (3)$$

where p or i represents a paper in the literature database.

Keyword-based Retrieval. Semantic-based retrieval returns a set of papers ($\mathbb{N}_1$) whose backgrounds align closely with the user’s query. Entities associated with these papers can play a crucial role in inspiring new ideas. To leverage

this, as shown in Figure 2, we extract entities from both the user’s query and the results of semantic-based retrieval. Specifically, keywords in the user’s query are extracted with prompt engineering (The prompt used for this task is detailed in the Appendix D.) with an LLM, while keywords for papers in $\mathbb{N}_1$ are retrieved directly from the literature database.

For a more thorough retrieval, we further expand the keyword set by employing a keyword-neighborhood-based approach. That is, for an keyword T_e in the current keyword set $\mathbb{E}_1$, any paper p that includes T_e will have its other keywords included in the candidate keyword set. While this may introduce redundant or even noisy keywords, we also employ a keyword filtering method and details are included in Appendix C.

The final expanded keyword set is denoted as $\mathbb{E}^{(u)}$. Keywords represent concepts that are most relevant to a paper’s topic. A paper is likely to be useful if it contains keywords matching those in $\mathbb{E}^{(u)}$. Consequently, for any keyword T_e in the set $\mathbb{E}^{(u)}$, as shown in Equation (4), we search the database for papers that also contain T_e , marking all retrieved papers as the set $\mathbb{N}_2$.

$$\mathbb{N}_2 = \{p | \exists T_e \in \mathbb{E}^{(u)} \wedge T_e \in T_b^{(p)}, p \in \mathbb{D}\}. \quad (4)$$

Co-occurrence-based Retrieval. In practice, we often encounter pairs of papers, p_1 and p_2 , which are neither similar in terms of keywords nor semantic themes but are frequently cited together. This co-citation suggests that researchers have identified a latent relationship between p_1 and p_2 in prior studies. To capture and leverage such implicit connections, we propose a literature retrieval method based on citation co-occurrence. As illustrated in Figure 2, for any paper p_1 that has already been retrieved, if p_2 is frequently cited alongside p_1 in other papers, p_2 will be included in our expanded literature retrieval set. Formally,

$$\mathbb{N}_3 = \{p_2 | p_1 \in (\mathbb{N}_1 \cup \mathbb{N}_2) \wedge \text{co-cite}(p_1, p_2)\}, \quad (5)$$

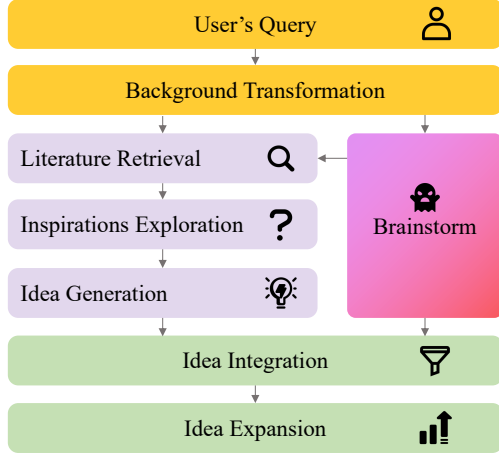


Figure 3. The pipeline of SciPIP for idea proposal.

where co-cite means p_1 and p_2 are often simultaneously cited by other papers. In practice, we select the $nCOCITE$ papers that are most frequently co-cited with each paper.

Finally, the whole retrieved papers can be represented as $\mathbb{N} = \mathbb{N}_1 \cup \mathbb{N}_2 \cup \mathbb{N}_3$.

Literature Clustering and Filtering. The SKC framework can retrieve hundreds of papers; however, not all of them will be practically referenced. For instance, when describing the background of a research field in a paper, researchers typically cite only a few representative papers rather than exhaustively referencing all relevant literature. To emulate this practice, we propose an idea-similarity-based clustering method.

Initially, each retrieved paper is treated as an individual cluster. Then, for any two clusters $\mathbb{A}$ and $\mathbb{B}$, they will be merged if the summary similarity between any pair of papers from $\mathbb{A}$ and $\mathbb{B}$ exceeds a predefined threshold τ . Here, the similarity between two papers is measured by the cosine similarity of their summary embeddings (*i.e.*, summary in Figure 1). After clustering, we set a total number of retrieved papers to $nPapers$ and evenly select the same number of papers from each cluster. This approach ensures diversity while reducing redundancy in the retrieved literature.

3.3. Dual-path Idea Generation

Researchers typically conceive new ideas through two distinct cognitive processes. On one hand, they engage in a thorough review of relevant literature, drawing inspiration from existing works to formulate novel ideas. On the other hand, they rely exclusively on their internal knowledge to brainstorm innovative concepts independently.

Inspired by these cognitive processes, we propose a dual-path framework for LLM-based idea generation that emu-

lates the reasoning patterns of researchers, as illustrated in Figure 3. Given a user’s query, the first path utilizes the literature retrieved in Section 3.2 to extract insights that may address the user’s challenges, subsequently generating ideas based on these insights. In contrast, the second path generates ideas without referencing external literature, relying entirely on the LLM’s internal knowledge. Finally, the ideas generated by both paths are integrated and refined, yielding a comprehensive and polished set of outputs.

3.3.1. BRAINSTORM-BASED GENERATION

This pathway generates ideas by relying exclusively on the LLM’s internal knowledge. However, we observe that the generation process is highly sensitive to the formulation of the user’s query. Specifically, the LLM may misinterpret the query if it is overly brief. To ensure more accurate brainstorming, we first transform the user’s query into a more detailed and structured format through prompt engineering. The LLM is instructed to carefully explain each technical term and supplement the query with additional context regarding why the stated problem constitutes a significant challenge. The specific prompt used for this transformation is provided in Appendix D. This process ensures that the transformed background explicitly includes a thorough description and a complete understanding of the query.

During the brainstorming phase, the LLM is provided with the transformed background and tasked with generating 3 to 4 ideas, each described in several sentences. The prompt used for this brainstorming process is detailed in Appendix D.

Following the brainstorming phase, we extract keywords from the generated ideas through prompt engineering. These keywords are then merged with those extracted from the background, providing a more robust foundation for subsequent literature retrieval and enhancing the overall thoroughness of the process.

3.3.2. RETRIEVAL BASED GENERATION

This pathway generates ideas by leveraging insights from retrieved literature. A critical challenge in this phase lies in effectively incorporating the retrieved literature into large language models (LLMs). Existing approaches often utilize titles or abstracts of retrieved papers as input. However, these components, along with the introduction sections, frequently lack the granularity required to comprehensively describe the underlying methodologies. Consequently, the insights derived by LLMs from such inputs are inherently limited. An alternative strategy involves inputting the entire paper to provide richer contextual information. Nevertheless, this approach introduces significant challenges, particularly in managing extremely long contexts, especially when a substantial number of papers are retrieved.

To address these limitations, we employ an LLM to summarize the methodology sections of the retrieved papers, aiming to generate concise yet informative descriptions. These descriptions are designed to be more detailed than the introduction but more succinct than the full methodology section, while excluding redundant content such as background information typically found in introductions. Empirical observations indicate that zero-shot instructions often fail to capture the core aspects of the methodology sections. To overcome this, we adopt a few-shot learning approach, where manually curated summaries of methodology sections are provided as demonstrations to guide the LLM in producing more accurate and focused summaries. The specific prompts used in this process are detailed in Appendix D.

Subsequently, the concise methodology summaries and the transformed background information are used as inputs to prompt the LLM to extract relevant inspirations. These inspirations focus on how the retrieved papers can address the challenges outlined in the transformed background. The extracted inspirations are further condensed, making them suitable as inputs for the idea generation phase. If the LLM determines that a particular paper offers no valuable insights, it is excluded from further consideration. Finally, the transformed background and the extracted inspirations are combined and fed into the LLM, which is then prompted to synthesize these insights and generate novel ideas. The specific prompts employed in this step are also provided in Appendix D.

Idea Integration and Expansion. Through an explicit dual-path generation process, two distinct groups of ideas are produced. These ideas are subsequently provided to the LLM for integration into approximately five coherent ideas. However, the integrated ideas are often brief and lack sufficient clarity. To address this, we employ prompt engineering to expand these ideas into more detailed versions. During this process, we observed that the quality, style, and level of detail in the expanded ideas are challenging to control solely through instructions. Therefore, we retrieve a “concise method” section from an existing paper as a demonstration and prompt the LLM to expand the proposed ideas accordingly. The prompts used for integration and expansion are detailed in Appendix D.

4. Experiments

4.1. Implementation Details

Literature Database. We curate a comprehensive dataset comprising 78,571 papers from nine prominent conferences in machine learning, natural language processing, and computer vision (ICML, NeurIPS, ICLR, ACL, EMNLP, NAACL, CVPR, ICCV, and ECCV) published over the past

Table 1. The subjective evaluation metrics of ideas quality. Higher means better.

Metric	Definition	Rating
Novelty	The idea is novel.	0 ~5
Clarity	The idea is clearly explained.	0 ~5
Feasibility	The idea will work (be effective) in my experience.	0 ~5
Relevance	The idea is relevant to the research background.	0 ~5
Helpfulness	The idea is helpful to researchers.	Yes / No

decade. The backgrounds, ideas, and other relevant information from these papers were summarized using GPT-4o. The summarized content was subsequently encoded using jina-embedding-v3 (Sturua et al., 2024) to facilitate efficient retrieval and analysis.

Literature Retrieval. For the retrieval process, we empirically set the hyperparameters as follows: $TopK = 5$, meaning that the semantics-based retrieval returns the top 5 most relevant papers, and $nCOCTE = 1$, indicating that co-occurrence retrieval adds one additional paper for each initially retrieved paper. The similarity threshold τ is set to 0.8, and $nPapers$ is fixed at 10 based on empirical validation. An ablation study is conducted to evaluate the impact of this hyper-parameter on the overall performance.

Test Data Construction. To address the challenges associated with manually constructing user’s query, we adopt a systematic approach by randomly selecting 34 papers from ACL 2024 and 25 papers from CVPR 2024. The research backgrounds from these papers are directly utilized as test cases. Importantly, the selected papers had not been pre-released on arXiv prior to their official publication, thereby mitigating the risk of data contamination¹.

Idea Proposals Evaluation. The evaluation of proposed ideas is conducted by a panel of 12 human researchers. They are graduate students or PhD candidates in the field of artificial intelligence, each with expertise in natural language processing or computer vision. All evaluators had published at least one first-author paper at top-tier academic conferences, ensuring a high level of domain expertise. They are tasked with assessing each proposed idea across five dimensions: novelty, clarity, feasibility, relevance, and helpfulness, as detailed in Table 1. Additionally, a pairwise comparison experiment was conducted, wherein evaluators were presented with two ideas generated under the same background and asked to select the one that performed better in terms of novelty, clarity, and other specified criteria.

¹The LLMs employed for idea generation (GPT-4o and GPT-4o-mini) have a data cutoff of October 2023.